

Suppression of Phase Noise Induced by Modulation Frequency Deviation in Fiber-Optic Gyroscopes

Xinyu Cao , Lanxin Zhu , Yanjun Chen , Fangshuo Shi , Wenbo Wang , Huimin Huang , and Zhengbin Li 

Abstract—The interferometric fiber-optic gyroscope (IFOG) is an instrument for rotational motion monitoring. In IFOGs, frequency modulation is required for noise suppression. However, when the modulation frequency is set based on a deviated eigen frequency, phase noise will increase in IFOGs. This paper proposes a novel second harmonic descent (SHD) algorithm for suppressing phase noise induced by modulation frequency deviation in IFOGs. This algorithm can obtain precise modulation frequency simply and quickly. The validity of the SHD algorithm has been verified in the static test of a IFOG. By using the SHD algorithm, the precision of measured modulation frequency can reach 0.01 Hz, and the corresponding relative error can reach 28 ppb ($1 \text{ ppb} = 1 \times 10^{-9}$). Besides, the modulation frequency obtained by the SHD algorithm can reduce the angular random walk (ARW) by 4.5-fold, which proves the effectiveness of the suppression of phase noise.

Index Terms—Fiber-optic gyroscope, modulation frequency, phase noise suppression, second harmonic descent algorithm.

I. INTRODUCTION

THE interferometer fiber-optic gyroscope (IFOG) is a sensor for rotational motion measurement [1]. Due to its advantages of solid-state configuration, high dynamic range and flat frequency response, it plays an important role in the fields of angular velocity measurement, geophysical exploration [2], rotational seismology [3] and gravity gradient measurement [4].

In IFOGs, signal detection requires modulation and demodulation. By modulating the rotational signal onto a high-frequency carrier, low-frequency noises such as $1/f$ electronic noise can be suppressed. The modulation frequency is typically set as the eigen frequency. This serves to avoid parasitic signals that result

from birefringence modulation and harmonic generation by the phase modulator, such as Rayleigh backscattering error [5]. For an IFOG with an eigen frequency of 176 kHz, when the applied eigen frequency deviates from the true value by 18 kHz, the noise level can be 10 times larger [6]. To enhance the measurement performance, increasing the diameter and length of the fiber-optic coil is a straightforward method. However, the thermal phase noise (TPN) affecting IFOG performance increases with length [4], [7]. To suppress TPN, high-order eigen frequency modulation is applied [8]. The modulation frequency is set at the eigen frequency and its odd harmonics, suppressing not only TPN, but also the thermal sensitivity of the IFOG [7]. However, when the high-order eigen frequency deviates from the exact value of 400 Hz, the above suppression effect on self-noise will be cancelled out [9]. As can be seen above, whether for the suppression of Rayleigh backscattering error or TPN, the deviation of modulation frequency, especially the value of eigen frequency, will result in an increase in the noise level of IFOG. Therefore, precise measurement of eigen frequency is very important.

Some schemes for the precise measurement of eigen frequency have been proposed. The most commonly used method is to measure the eigen frequency by square wave feedback. When the feedback duty cycle of the square wave reaches 50%, the frequency of the square wave corresponds to half of the exact value of the eigen frequency. The method can achieve a precision of 10 Hz [10]. Setting the feedback waveform as a combination of sawtooth and square waves can increase the precision up to 0.1 Hz [11]. Besides, the output of the IFOG can be used to measure the eigen frequency indirectly. When the square wave modulation frequency reaches the even-order eigen frequency, the output approaches 0. This method achieves a precision of 75 Hz [12]. By adjusting modulation phase depth and using over-bias modulation, the precision can be further improved to 5 Hz [13]. However, these methods all rely on frequency sweeping. Thus to achieve higher precision, they consume a larger amount of computational resources. Moreover, these methods are particularly sensitive to the signal-to-noise ratio (SNR) of the acquired signals, becoming less reliable when SNR is poor. Additionally, these methods are primarily designed for eigen frequency modulation. When high-order eigen frequency modulation is employed, the error accumulates and ultimately contributes to a decline in the performance of the IFOG.

In this study, we propose a novel second harmonic descent (SHD) algorithm for precise eigen frequency measurement. By leveraging the convexity and symmetry of the second harmonic

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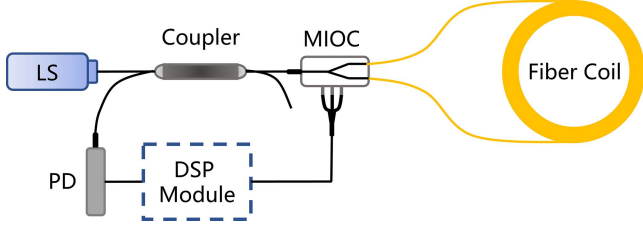


Fig. 1. The configuration of the IFOG.

of the IFOG signal, it can help to converge the high-order eigen frequency to the extreme points directly. Meanwhile, by taking advantage of the monotonicity of the convergence process, the SHD algorithm can effectively obtain the eigen frequency with high precision under the condition of low SNR. We validate this algorithm on a IFOG. The experimental results show that by using the proposed SHD algorithm, the measurement precision of high-order eigen frequency can reach 0.01 Hz, and the corresponding relative error can reach 28 ppb. Compared with setting the high-order eigen frequency based on the conventional square wave feedback method, setting the high-order eigen frequency based on the proposed SHD algorithm can improve ARW by 4.5×times. This proves the effectiveness of the SHD algorithm in suppressing the phase noise induced by the modulation frequency deviation.

II. ANALYSIS OF MODULATION FREQUENCY DEVIATION

A. Demodulation of the IFOG

The configuration of the IFOG is shown in Fig. 1. Light from a broadband amplified spontaneous emission (ASE) light source is transmitted through a coupler. Then it is split into two paths by a multi-function integrated optic chip (MIOC), propagating in both clockwise and counterclockwise directions within the fiber-optic coil. After traversing the coil, the light beams are recombined by the MIOC and coupler, and then detected by a photodetector (PD). The digital signal processing (DSP) module applies sinusoidal modulation to the MIOC and demodulates the optical interference signals received by the PD, ultimately obtaining the angular velocity measured by the IFOG. For that the modulation involves no feedback control, it is an open-loop IFOG system. For a fiber-optic coil with a length L and diameter D , when the rotational angular velocity is Ω , the induced Sagnac phase is expressed as [14]:

$$\phi_S = \frac{2\pi LD}{\lambda c} \Omega, \quad (1)$$

where λ is the central wavelength of light, and c is the light speed in vacuum. The sinusoidal modulation signal applied with a frequency of f_m and an amplitude of φ_0 can be represented as:

$$\varphi(t) = \varphi_0 \sin(2\pi f_m t). \quad (2)$$

The light is phase-modulated in both directions, and the accumulated phase difference is written as:

$$\Delta\varphi(t) = \varphi(t) - \varphi(t - \tau) \triangleq \phi_m \cos\left[2\pi f_m \left(t - \frac{\tau}{2}\right)\right], \quad (3)$$

where $\tau = L/v$ is the transit time of the coil, $v = c/n_{\text{eff}}$ is the relative propagation speed of light in the optical fiber and n_{eff} is the effective refractive index. $\phi_m = 2\varphi_0 \sin(\frac{\pi f_m}{2f_e})$ is defined as the modulation depth, and f_e is the eigen frequency of the fiber-optic coil, defined as:

$$f_e = \frac{c}{2n_{\text{eff}}L}. \quad (4)$$

Thus, for the open-loop IFOG, the detected signal can be expressed as [15]:

$$V_D(t) = G_0 I_0 \{1 + \cos[\phi_S + \Delta\varphi(t)]\}, \quad (5)$$

where G_0 is the gain of the detection system, and I_0 is the intensity of the light source. Expand (5) using the Jacobi-Anger expansion and it can be given as:

$$\begin{aligned} V_D(t) = G_0 I_0 & \left\{ 1 + \cos(\phi_S) \left[J_0(\phi_m) \right. \right. \\ & + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(\phi_m) \cos \left[2n \cdot 2\pi f_m \left(t - \frac{\tau}{2} \right) \right] \\ & + \sin(\phi_S) \cdot 2 \sum_{n=1}^{\infty} (-1)^n J_{2n-1}(\phi_m) \\ & \left. \left. \times \cos \left[(2n-1) \cdot 2\pi f_m \left(t - \frac{\tau}{2} \right) \right] \right] \right\}, \quad (6) \end{aligned}$$

where J_n represents the n -th order of the first kind Bessel function. From (6), it can be seen that the expanded signal includes a DC component, odd harmonic components and even harmonic components. The amplitudes of these harmonics can be obtained through Fourier transform or coherent detection. Take the 1st, 2nd, and 4th harmonic amplitudes, denoted as:

$$\begin{aligned} S_1 &= 2G_0 I_0 J_1(\phi_m) \sin(\phi_S), \\ S_2 &= 2G_0 I_0 J_2(\phi_m) \cos(\phi_S), \\ S_4 &= 2G_0 I_0 J_4(\phi_m) \cos(\phi_S). \end{aligned} \quad (7)$$

According to (7), the modulation depth ϕ_m is determined by S_2/S_4 . The Sagnac phase shift ϕ_S caused by rotational motion can be calculated by S_1/S_2 . Subsequently, substitute the calculated ϕ_m into the following expression and the rotational velocity can be solved.

$$\Omega = \frac{\lambda c}{2\pi LD} \cdot \tan^{-1} \left[\frac{S_1}{S_2} \cdot \frac{J_2(\phi_m)}{J_1(\phi_m)} \right]. \quad (8)$$

B. Noise Induced by Modulation Frequency Deviation

As stated in Section I, for giant IFOGs, the performance is primarily limited by TPN. TPN comes from random thermal motion, which changes the effective refractive index of optical fiber [4]. TPN imposes an additional interference phase shift ϕ_T in the fiber-optic coil. After modulation and demodulation, the root power spectral density (PSD) of ϕ_T is denoted as $S_{\phi_T}(f)$,

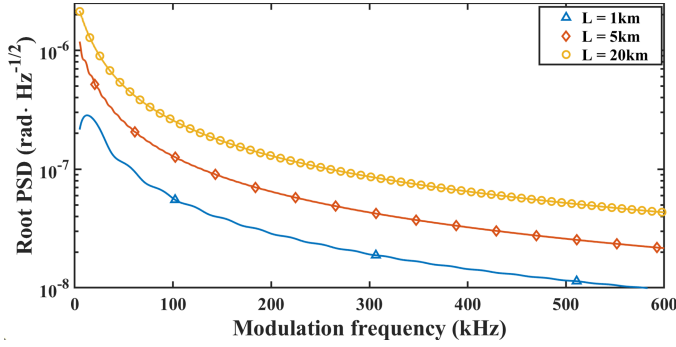


Fig. 2. Root PSD of $S_{\phi_T}(f)$ under different fiber length L and modulation frequency f_m . The triangles, diamonds and circles represent the eigen frequency and its odd harmonics of the 1 km, 5 km and 20 km fiber-optic coil, respectively.

shown in (9) [7].

$$S_{\phi_T}(f) = \frac{1}{|J_1(\phi_m)|} \left\{ \sum_{n=0}^{\infty} J_{2n+1}^2(\phi_m) \left\{ \langle \Delta\phi_N^2[(2n+2)f_m] \rangle + \langle \Delta\phi_N^2(-2nf_m) \rangle \right\} \right\}^{1/2},$$

$$\langle \Delta\phi_N^2(f) \rangle = \frac{k_B T^2 L}{\kappa \lambda^2} \left(\frac{dn_{\text{eff}}}{dT} + n_{\text{eff}} \alpha \right)^2 \ln \left[\frac{\left(\frac{2}{W_0} \right)^4 + \left(\frac{2\pi f}{D_0} \right)^2}{\left(\frac{4.81}{d} \right)^4 + \left(\frac{2\pi f}{D_0} \right)^2} \right] \times \left[1 - \text{sinc} \left(\frac{2\pi f L n_{\text{eff}}}{c} \right) \right], \quad (9)$$

where k_B is the Boltzmann constant, κ is the thermal conductivity, α is the linear thermal expansion coefficient, W_0 is the mode field radius of optical fiber, D_0 is the thermal diffusivity, d is the cladding diameter. It can be seen that the root PSD of TPN is directly affected by fiber length L and the modulation frequency f_m . To further investigate the relationship between L , f_m , and $S_{\phi_T}(f)$, a simulation is conducted as shown in Fig. 2. Considering that the value of the fiber length L in experiment is about 20 km, and $L = 1$ km, 5 km, 20 km are commonly used fiber length [1], [3], [15], [16], [17], these three values are selected for simulation.

From Fig. 2 we can see that it is necessary to use high-order eigen frequency modulation to shift the operating point of the gyro to a frequency band with lower TPN. The mentioned high-order eigen frequencies are actually odd-order eigen frequencies, to ensure consistency in Rayleigh backscattering noise [16]. Although the eigen frequency of the fiber-optic coil can be calculated by (4), the fiber length L provided by manufacturers is typically not accurate enough. Moreover, fiber length can vary with changes in temperature and humidity. Therefore, the eigen frequency calculated in this way is not credible. When biased modulation frequency is applied, it will deteriorate the performance of the IFOG, primarily due to parasitic phase error caused by Rayleigh backscattering. Denote backscattered light and the input light as I_{bs} and I_0 , respectively. The relationship

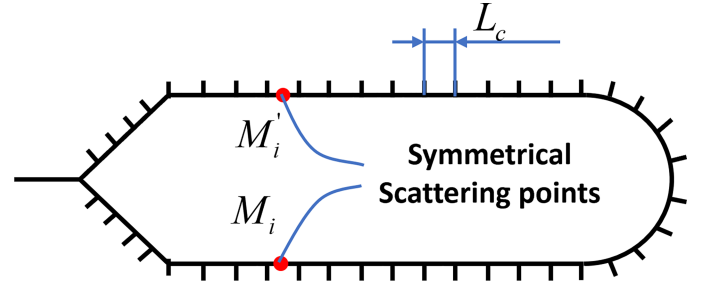


Fig. 3. Schematic diagram of symmetrical Rayleigh backscattering points in the fiber-optic coil.

between I_{bs} and I_0 is given by (10):

$$\frac{I_{bs}}{I_0} = \left(1 - 10^{-\alpha_R L_c / 10} \right) \alpha_{bs} \triangleq \eta_{bs}, \quad (10)$$

where α_R is the fiber attenuation coefficient caused by Rayleigh backscattering, with a typical value of 0.2 dB/km. α_{bs} is the backscattering coefficient, with a typical value of 10^{-3} . L_c is the coherence length of the light source. Rayleigh backscattering causes not only a phase noise, but also an additional second-order intensity noise. As shown in Fig. 3, the backscattering along the coil is produced by several groups of backscattering points M_i , each corresponding to a length of L_c , totaling $N = L/L_c$ backscattering points. Each pair of symmetrical backscattering points will interfere, but only a pair within one coherence length at the midpoint of the coil will interfere with the main wave. For other backscattering points, there is only a simple intensity accumulation, resulting in intensity noise. Since sinusoidal modulation is used, according to (3), the phase difference for the i -th pair of symmetrical points can be noted as in (11):

$$\Delta\phi_i = 4\varphi_0 \cos \left(2\pi f_m \cdot \frac{i}{N} \tau \right) \sin \left[2\pi f_m \left(t - \frac{i}{N} \tau \right) \right] + \varphi_i. \quad (11)$$

In (11), $\frac{i}{N} \tau$ represents the time when backscattering occurs, while φ_i represents the random phase difference between the symmetrical backscattering points M_i and M_i' . The intensity of the interference can be expressed as:

$$I_{bs,i} = \eta_{bs} I_0 [(1 + \cos(\Delta\phi_i))]. \quad (12)$$

Denote that $\phi_{m,i} = 4\varphi_0 \cos(2\pi f_m \cdot \frac{i}{N} \tau)$, and expand (11) by the Jacobi-Anger expansion. It gives that:

$$I_{bs,i} = \eta_{bs} I_0 \left\{ 1 + \cos(\varphi_i) \left[J_0(\phi_{m,i}) + 2 \sum_{n=1}^{\infty} J_{2n}(\phi_{m,i}) \cos \left[2n \cdot 2\pi f_m \left(t - \frac{i}{N} \tau \right) \right] + \sin(\varphi_i) \cdot 2 \sum_{n=1}^{\infty} J_{2n-1}(\phi_{m,i}) \times \sin \left[(2n-1) \cdot 2\pi f_m \left(t - \frac{i}{N} \tau \right) \right] \right] \right\}. \quad (13)$$

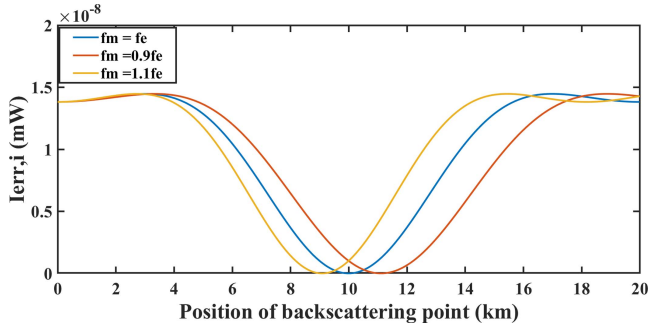


Fig. 4. Intensity of second-order Rayleigh backscattering error caused by backscattering points at different positions in the fiber-optic coil under various f_m .

The modulation signal is a sinusoidal signal with a frequency f_m given by (2), so that only the first harmonic in (13) contributes to the error. After phase-sensitive detection, the amplitude of the error signal is written as:

$$I_{err,i} = 2\eta_{bs}I_0 \left| \sin(\varphi_i) J_1(\phi_{m,i}) \cos\left(2\pi f_m \frac{i}{N} \tau\right) \right|. \quad (14)$$

The intensity of the light source is set as 100 mW. When f_m deviates from the eigen frequency f_e , the intensity of second-order Rayleigh backscattering noise caused by backscattering points at different positions in the 20-km fiber-optic coil is given by Fig. 4.

From Fig. 4, it can be seen that as f_m deviates, the distribution of Rayleigh backscattering points changes, which in turn affects the intensity noise caused by the second-order backscattering, denoted as I_{err} . Since φ_i is random and uniformly distributed within the range $[0, 2\pi]$ [16], the mean value of $\sin^2(\varphi_i)$ is $1/2$. For that each group of symmetrical points is independent, I_{err} can be written as the sum of the root mean squares of the errors of each group of points, denoted as:

$$I_{err} = \sqrt{2}\eta_{bs}I_0 \sqrt{\sum_{i=1}^N \left[J_1(\phi_{m,i}) \cos\left(2\pi f_m \frac{i}{N} \tau\right) \right]^2}. \quad (15)$$

Equation (15) shows the expression of intensity noise caused by second-order Rayleigh backscattering. For the output of the IFOG in (5), this results in an additional intensity superimposed directly on the original output intensity, leading to amplitude change of the output signal. A simulation is conducted to demonstrate how I_{err} changes with the modulation frequency f_m , with different fiber length $L = 5$ km and $L = 20$ km. Intensity of light source is set as 100 mW, and f_m is varied from 0 Hz to 100 kHz.

Fig. 5 shows that the intensity of I_{err} changes minimally in amplitude as the modulation frequency increases and will gradually converge. While considering different fiber lengths, longer fiber implies more groups of symmetric backscattering points, resulting in larger total intensity noise. In fact, the mean value of I_{err} is proportional to the square root of the fiber length $\sqrt{L}$. When considering the phase error caused by I_{err} , the basic output of the IFOG can be written as $I_D = I_0[1 + \cos(\phi_S)] + I_{err}$. Assuming that the IFOG operates with a phase offset of $\pi/2$ and only detects the Sagnac phase shift caused by the rotation

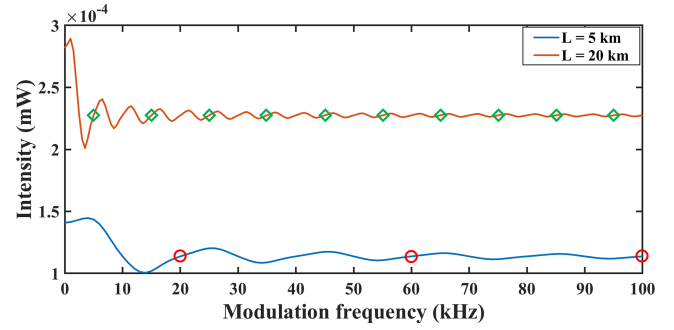


Fig. 5. Intensity of the second-order Rayleigh backscattering noise I_{err} at different modulation frequency, with fiber length $L = 5$ km and $L = 20$ km. The circles and diamonds represent the eigen frequency and its odd harmonics of the 5 km and 20 km fiber-optic coil, respectively.

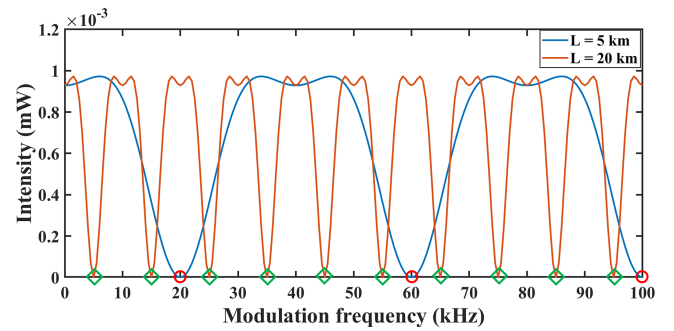


Fig. 6. Intensity of the phase noise caused by Rayleigh backscattering $I_{err,ph}$ at different modulation frequency, with fiber length $L = 5$ km and $L = 20$ km. The circles and diamonds represent the eigen frequency and its odd harmonics of the 5 km and 20 km fiber-optic coil, respectively.

of the Earth, it gives that:

$$\frac{I_D}{I_0} \approx \phi_S + \frac{I_{err}}{I_0} \triangleq \phi_S + \phi_{bs,I}, \quad (16)$$

which implies that the phase error caused by the intensity noise of second-order Rayleigh backscattering in a 20 km long fiber-optic coil can be denoted as:

$$\phi_{bs,I} = \frac{I_{err}}{I_0} \approx 2.3 \times 10^{-6} \text{ rad}. \quad (17)$$

Besides the intensity noise caused by second-order Rayleigh backscattering, there is also phase noise in the fiber-optic coil resulting from the interference between backscattering light and the main light. For a set of interference points at $i = N/2$, the interference between them and the main light constitutes the phase noise. According to (11)–(13), it can be denoted as $I_{err,ph}$:

$$I_{err,ph} = \sqrt{2} \cdot \sqrt{\eta_{bs}I_0} \left| J_1\left(\phi_{m,\frac{N}{2}}\right) \cos\left(\frac{\pi f_m}{2f_e}\right) \right|. \quad (18)$$

When the modulation frequency f_m varies from 0 Hz to 100 kHz, the intensity $I_{err,ph}$ caused by phase noise of Rayleigh backscattering in fiber-optic coil with the length of $L = 5$ km and $L = 20$ km is shown in Fig. 6. The points in red circles correspond to the eigen frequency and the odd harmonics of the 5 km fiber-optic coil, and the green diamonds represents the eigen frequency

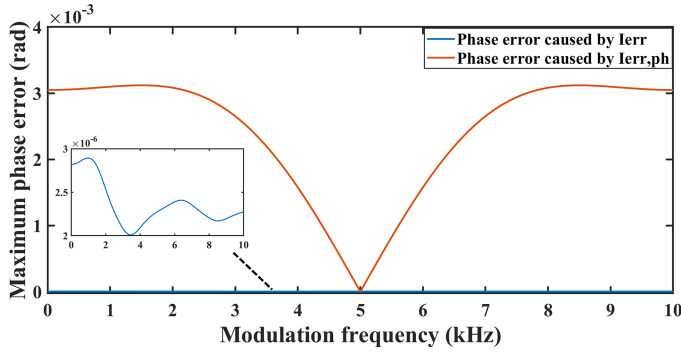


Fig. 7. The maximum phase error caused by I_{err} and $I_{err,ph}$ in a 20 km long fiber-optic coil when the modulation frequency f_m deviates from the eigen frequency f_e in one period.

and the odd harmonics of the 20 km fiber-optic coil. It can be seen from Fig. 6 that as the modulation frequency increases, the intensity caused by phase noise $I_{err,ph}$ exhibits periodic changes with f_m . $I_{err,ph}$ can be completely suppressed only when f_m is at the eigen frequency and its odd harmonic points, and rises rapidly when f_m deviates from these points. Moreover, the intensity of the phase noise is unrelated to the fiber length because only the pair of backscattering points in the middle of the coil interferes with the main light.

While considering the effect of $I_{err,ph}$, the amplitude ratio of the backscattered light to the main signal light is an estimate of this phase error [6], expressed as:

$$\phi_{bs,ph} \approx \frac{E_{err,ph}}{E_0} = \sqrt{\frac{I_{err,ph}}{I_0}}. \quad (19)$$

According to (19), the maximum phase noise caused by $I_{err,ph}$ is $\phi_{bs,ph} \approx 3.3 \times 10^{-3}$ rad, which has much bigger impact than I_{err} . To better evaluate the effect of I_{err} and $I_{err,ph}$ in a 20 km long fiber-optic coil, the maximum phase error of I_{err} and $I_{err,ph}$ when f_m deviates from f_e in one period is given in Fig. 7 separately:

From Fig. 7, the maximum phase error caused by the phase noise is much greater than that caused by the intensity noise. In addition, $I_{err,ph}$ can be completely suppressed when f_m is at the eigen frequency. Combined with Figs. 5 and 6, for the noise caused by Rayleigh backscattering, including intensity noise and phase noise, the modulation frequency should be set at the eigen frequency and its odd harmonics. This can completely suppress the dominant phase noise, while the intensity noise is an extremely small quantity that is almost not affected by the high-order modulation frequency, which can be eliminated by compensation or ignored directly.

However, when the modulation frequency deviates from the true value, the suppression effect caused by the eigen frequency modulation or its odd harmonic frequency modulation will be degraded. When the modulation frequency deviates from the eigen frequency 5 kHz by 1%, i.e., 50 Hz, the maximum phase error caused by phase noise due to Rayleigh backscattering reaches 8.4×10^{-5} rad. When the modulation frequency deviates by 2%, i.e., 100 Hz, this phase error rapidly increases to 1.7×10^{-4} rad.

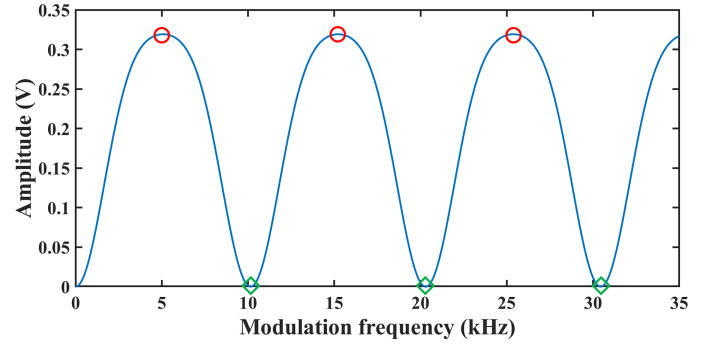


Fig. 8. Amplitude of the second harmonic of the IFOG signal. The red circles mark the odd frequency points of the eigen frequency, and the green diamonds mark the even ones.

As the deviation accumulates, when the modulation frequency deviates by 10% from the eigen frequency, with a deviation frequency of 500 Hz, the corresponding maximum phase error reaches 8.2×10^{-4} rad. This is non-negligible for the Sagnac phase shift of approximately 2.5×10^{-2} rad in such a fiber-optic coil and can cause degradation in the performance of the IFOG.

In summary, TPN and Rayleigh backscattering can be suppressed by high-order eigen frequency modulation. However, when the modulation frequency deviates from the odd harmonics of the eigen frequency, the phase noise caused by the deviation of modulation frequency will significantly affect the performance of the IFOG. Therefore, there is an urgent need for a highly precise method to determine the eigen frequency.

III. THE PRINCIPLE OF SUPPRESSING THE EFFECT INDUCED BY MODULATION FREQUENCY DEVIATION

A. Characteristics of Harmonics of IFOG Signal

When the IFOG is in stationary conditions, the Sagnac phase shift ϕ_s is caused by the rotation of the Earth. Nevertheless, the corresponding first harmonic value remains too small to detect [18]. To achieve the best SNR, the modulation depth ϕ_m is set at 2.7 rd [17]. Under this modulation depth and Sagnac phase shift, the second harmonic has the largest amplitude among all detected harmonics. According to (6), the amplitude of the second harmonic can be expressed as:

$$A = 2G_0 I_0 J_2(\phi_m) \cos(\phi_s). \quad (20)$$

To quantitatively evaluate this, a numerical simulation is performed. The main typical parameters in the simulation are: fiber length $L = 20652$ m, gyro diameter $D = 2.05$ m, fiber effective refractive index $n_{eff} = 1.455$, the central wavelength of light source $\lambda = 1550$ nm, and the eigen frequency $f_e = 5073.94$ Hz.

The relationship between the amplitude of the second harmonic and the modulation frequency f_m is shown in Fig. 8. The red circles mark the odd frequency points of the eigen frequency, and the green diamonds mark the even ones. According to (20) and Fig. 8, it is suggested that the amplitude of the second harmonic reaches its minimum value 0 at a modulation frequency f_m equal to $2kf_e$. Additionally, within the specified range of $((2k-1)f_e, (2k+1)f_e)$, Fig. 8 demonstrates symmetry about

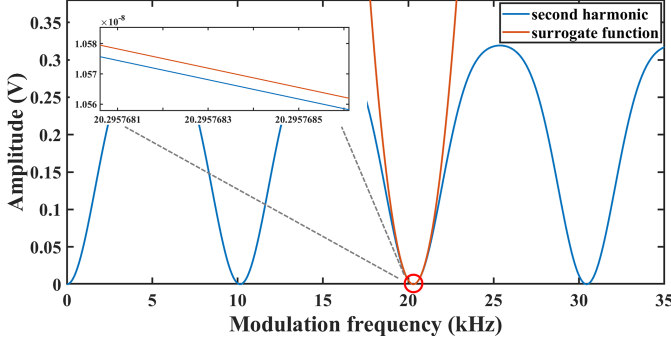


Fig. 9. Comparison of second harmonic and the surrogate function of second harmonic at $f_m = 4f_e$.

the point $2kf_e$. In fact, when the amplitude of the second harmonic is 0, the modulation depth ϕ_m also reaches 0 [19]. Therefore, by finding the minimum point corresponding to the modulation frequency f_m in the figure, f_e can be determined by $f_e = \frac{f_m}{2k}$. At this point, the second-order Bessel function can be approximated as [20]:

$$J_2(\phi_m) = \frac{1}{\Gamma(3)} \left(\frac{\phi_m}{2} \right)^2 - O(\phi_m^4). \quad (21)$$

In (21), $\Gamma(n) = (n-1)!$, and $O(\phi_m^4)$ is the fourth-order infinitesimal of ϕ_m , which can be neglected. By substituting (21) into (20), it can be obtained that near the point $f_m = 2kf_e$, the expression of the second harmonic can be approximately written in the form of a quadratic function as:

$$\tilde{A} = \frac{1}{4} G_0 I_0 \phi_m^2. \quad (22)$$

As f_m approaches $2kf_e$, (22) can be further simplified to:

$$\tilde{A} = \frac{G_0 I_0 \phi_0^2 \pi^2}{4 f_e^2} f_m^2. \quad (23)$$

Thus, when the modulation frequency f_m is close to $2kf_e$, the second harmonic can be approximately modeled as a quadratic function of f_m . This model theoretically simplifies the difficulty of finding the harmonic multiples of the eigen frequency. During the process of transitioning from A to $\tilde{A}$, each step in the approximation procedure amplifies the original value of A . Hence, there is:

$$\tilde{A} \geq A. \quad (24)$$

In fact, $\tilde{A}$ is strictly greater than A due to the presence of ϕ_s and higher order infinite minor terms in the second harmonic. Define $\tilde{A}(f)$ as the surrogate function of second harmonic of the IFOG signal, denoted as:

$$\tilde{A}(f) = \frac{G_0 I_0 \phi_0^2 \pi^2}{4 \hat{f}_e^2} f^2, \quad (25)$$

where $\hat{f}_e$ is an estimation for f_e . Assuming that $\hat{f}_e = f_e$, the comparison between the second harmonic A and the surrogate function $\tilde{A}(f - 4f_e)$ at $f_m = 4f_e$ is shown in Fig. 9.

Fig. 9 shows that the surrogate function is the upper bound of the second harmonic of the IFOG signal. Consequently,

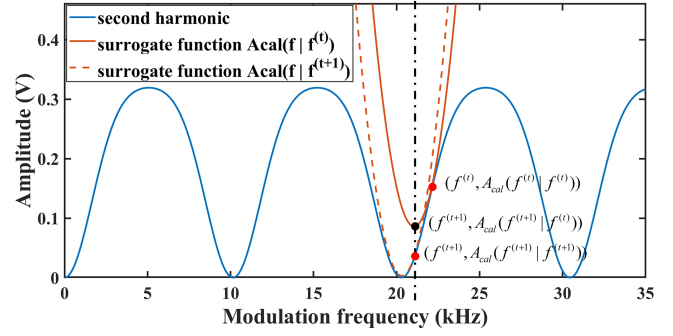


Fig. 10. The iteration process of SHD.

searching for the minimum point of the second harmonic can be transformed into finding the minimum point of $\tilde{A}(f)$. This forms the implementation basis for the subsequent SHD algorithm.

B. Second Harmonic Descent Algorithm

To identify the eigen frequency of a fiber-optic coil, we can sweep the modulation frequency f_m within a certain frequency band and directly locating the f_m at where the amplitude of the second harmonic becomes zero. At this point, f_m corresponds to an even multiple of the eigen frequency f_e . By combining this method with a rough estimation of the fiber-optic coil length, the precise value of the eigen frequency f_e and the exact length of the fiber can be accurately determined. However, sweeping the modulation frequency is cumbersome as it requires sweeping across multiple frequency points and comparing the amplitudes of the second harmonic obtained. Additionally, fine adjustments to f_m are needed when the amplitude approaches zero, adding to the complexity. Moreover, when the amplitude is small, the measured average and standard deviation are of the same magnitude, significantly impacting the precision of identifying the ideal multiple points of the eigen frequency.

Based on the aforementioned issues, we propose the second harmonic descent (SHD) algorithm. This algorithm enables rapid and precise identification of the extreme points of the second harmonic, with lower susceptibility to noise interference.

The principle of the SHD algorithm is based on the analysis of the second harmonic amplitude A and the surrogate function $\tilde{A}(f)$ as described in Section III-A. Construct a surrogate function for calculation $A_{cal}(f)$ of the form $\tilde{A}(f)$, and the search for the minimum point of A can be transformed into the search for the minimum point of $A_{cal}(f)$. This surrogate function $A_{cal}(f)$ always acts as an upper bound to the amplitude of the second harmonic of the IFOG signal. By minimizing $A_{cal}(f)$ in each iteration, this method allows for a gradual approximation to the minimal value of the original amplitude function A . The iteration process can be intuitively seen from Fig. 10 and the specific steps of the SHD algorithm are as follows:

1) *Step 1:* Select a frequency $f^{(0)}$ as the starting point. The choice of $f^{(0)}$ is arbitrary.

2) *Step 2:* Denote the current estimation for the modulation frequency f_m as $f^{(t)}$. Construct the surrogate function $A_{cal}(f | f^{(t)})$ for the second harmonic function $A(f)$, ensuring

that: $A_{cal}(f|f^{(t)}) \geq A(f)$, and $A_{cal}(f^{(t)}|f^{(t)}) = A(f^{(t)})$. In the SHD algorithm, the constructed function is as follows:

$$A_{cal}(f|f^{(t)}) = \frac{G_0 I_0 \varphi_0^2 \pi^2}{4 \hat{f}_e^2} (f - a)^2 + b, \quad (26)$$

where $\hat{f}_e$ is an estimation for f_e . a, b are constants that satisfy the following relationship:

$$\begin{aligned} A_{cal}(f^{(t)}|f^{(t)}) &= A(f^{(t)}), \\ A'_{cal}(f^{(t)}|f^{(t)}) &= A'(f^{(t)}). \end{aligned} \quad (27)$$

3) *Step 3*: Determine the next frequency point $f^{(t+1)}$ to minimize $A_{cal}(f)$.

$$f^{(t+1)} = \arg \min_f A_{cal}(f|f^{(t)}). \quad (28)$$

Based on (26), the solution is:

$$f^{(t+1)} = f^{(t)} - \frac{2 \hat{f}_e^2}{G_0 I_0 \varphi_0^2 \pi^2} A'(f^{(t)}). \quad (29)$$

4) *Step 4*: Repeat steps 2 and 3 until the difference between $f^{(t+1)}$ and $f^{(t)}$ is less than the error threshold ε_f , at which point the iteration stops. The corresponding $f^{(t+1)}$ is an even multiple of the eigen frequency f_e .

C. Error Analysis

In the SHD process, stopping the iteration depends on the error threshold ε_f . This is also the primary source of measurement error. The measured modulation frequency $\tilde{f}_m$ is denoted as:

$$\tilde{f}_m = 2k f_e + \varepsilon_f. \quad (30)$$

From (30), the relationship between the measured $\tilde{f}_e$ and the actual eigen frequency f_e can be directly written out as:

$$\tilde{f}_e = \left(1 + \frac{\varepsilon_f}{2k f_e}\right) f_e. \quad (31)$$

According to (31), as ε_f is typically a fixed value, the measurement error of $\tilde{f}_e$ is determined by $2k f_e$. For that the maximum value of $2k f_e$ depends on the maximum sampling rate of the system, $\frac{\varepsilon_f}{2k f_e}$ is fixed and can be denoted as ε_L . Fiber length error ΔL is written as:

$$\Delta L = \frac{\varepsilon_L}{1 + \varepsilon_L} L. \quad (32)$$

From (30)–(32) it can be seen that by using this algorithm, the measurement error of eigen frequency is determined, and measurement error of fiber length is proportional to the fiber length itself. It ultimately depends on the error threshold ε_f .

From the previous analysis, it is preferable to set ε_f as small as possible. Its limit primarily depends on the voltage detection limit of the measurement system. Denote the minimum detectable voltage as V_e , and the following relationship needs to be satisfied:

$$A_{cal}(f^{(t)}) - A_{cal}(f^{(t+1)}) > V_e. \quad (33)$$

The minimum value of $A_{cal}(f^{(t)}) - A_{cal}(f^{(t+1)})$ is achieved in the final iteration step of the SHD algorithm, and it can be

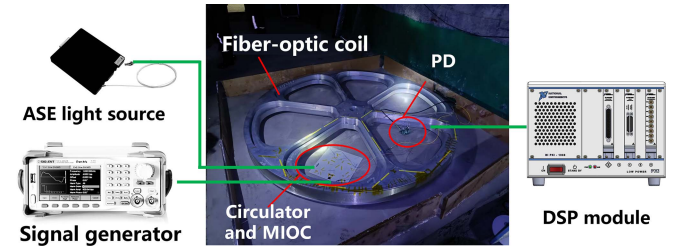


Fig. 11. IFOG configuration and photographs.

estimated as:

$$A_{cal}(f^{(t)}) - A_{cal}(f^{(t+1)}) \approx \frac{k G_0 I_0 \varphi_0^2 \pi^2}{\hat{f}_e} \varepsilon_f. \quad (34)$$

Therefore, the limit of ε_f can be expressed as:

$$\varepsilon_f > \frac{V_e \hat{f}_e}{k G_0 I_0 \varphi_0^2 \pi^2}. \quad (35)$$

Equation (35) shows the primary limitation of the SHD algorithm. During the experiment, a 24-bit analog-to-digital converter with an input range of -5V to +5V is used, thus its minimum voltage resolution is approximately calculated as $V_e = 6 \times 10^{-7}$ V. For a 20 km fiber-optic coil, its eigen frequency can be estimated as $\hat{f}_e = 5000$ Hz. Due to the limitations of the measurement system, it can be estimated that $k = 30$. Therefore, the range of ε_f can be determined as:

$$\varepsilon_f > 10^{-5} \text{ Hz}. \quad (36)$$

Equation (36) indicates that the eigen frequency and the fiber length measured by the SHD algorithm are highly precise. This will be elaborated in detail in the following experiment section.

IV. EXPERIMENTS AND RESULTS

The verification of the SHD algorithm is conducted on a giant IFOG. The test system is shown in Fig. 11, adopting the minimum configuration illustrated in Fig. 1. The light source is an ASE broadband light source. A polarization-maintaining circulator is added between the light source and the MIOC to ensure the polarization state of light during transmission. The MIOC is responsible for polarizing filtering and splitting the light, and the sinusoidal modulation signal generated by the signal generator is also applied to the light signal at the MIOC. After propagation in the giant fiber-optic coil, the light signal is combined by the MIOC, detected by the PD, and converted into a digital electrical signal. Finally, the actual signal detected by giant IFOG is obtained by processing and demodulating the electrical signal.

The main parameters in the experiment are shown in Table I. The fiber-optic coil is constructed by polarization-maintaining fiber and uses an aluminum alloy frame as its overall support structure. The nominal length of the fiber is 20682 m, and the diameter of the coil is 2.05 m. During the winding process of the fiber-optic coil, a quadrupole symmetric method is adopted, which has greatly reduced phase noise caused by Shupe error [1]. The PDs used are PIN photodiodes, with a

TABLE I
VALUES OF THE PARAMETERS IN EXPERIMENT

Notation	Property	Value
λ	Central wavelength of light	1.55×10^{-6} m
n_{eff}	Effective refractive index	1.455
L	Nominal fiber length	20682 m
D	Diameter of the fiber-optic coil	2.05 m
$\tilde{f}_e$	Obtained eigen frequency	5050 Hz
ϕ_m	Modulation depth	2.7 rad

TABLE II
ITERATION PROCESS OF SHD

Iteration step	$f^{(t)}$	$f^{(t+1)}$	ε_f
0	356000.000	355299.067	700.933
1	355299.067	355185.060	114.007
2	355185.060	355181.354	3.706
3	355181.354	355181.245	0.109
4	355181.245	355181.242	0.003

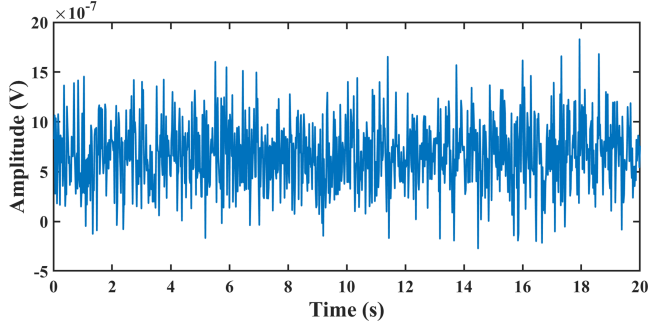


Fig. 12. The amplitude of second harmonic at $2\tilde{f}_e$ by conventional square wave feedback method.

sensitivity of 0.90 mA/mW. After the conversion from light to electricity, signal digitization and demodulation are carried out on a PXI-5922 oscilloscope, with a sampling rate of 4 MHz.

The obtained eigen frequency $\tilde{f}_e$ in the table is determined by conventional square wave feedback method [10]. This method adjusts the frequency of the square wave signal applied to the MIOC, and make the duty cycle of the square wave output to be 50%. At this time, the modulation frequency is half of the eigen frequency. According to $\tilde{f}_e$, the length of the coil is calculated to be 20400.290 m, which differs significantly from the nominal length of 20682 m. The amplitude of second harmonic is measured at the frequency of $2\tilde{f}_e = 10100$ Hz, as shown in Fig. 12. The mean value is 6.77×10^{-7} V and the standard deviation is 3.37×10^{-7} V. This means that the coefficient of variation of the second harmonic amplitude almost reaches 50%, making the result highly unreliable. By contrast, the SHD algorithm can determine whether a data point is reliable based on the monotonic trend of $f^{(t)}$ during prior iterations, even in low SNR conditions. Additionally, even if the mean value is unreliable, as long as it maintains the monotonicity of the convergence process, it can still iterate and converge to the threshold ε_f . These properties are jointly determined by the symmetry of the second harmonic A within one period, the symmetry of the surrogate function A_{cal} , and the fact that A_{cal} is an upper bound of A .

In implementing the SHD algorithm, it is first necessary, as mentioned in Section III-A, to determine the surrogate function $A_{cal}(f)$. As shown in Fig. 10, $A_{cal}(f)$ must always remain above the original second harmonic of the IFOG signal within one period. Therefore, when determining parameters, the quadratic term coefficient should be slightly overestimated,

which is reflected in slightly underestimate the eigen frequency, denoted as $\hat{f}_e = 5000$ Hz. Based on the detection limit of the analog-to-digital conversion system, set the modulation frequency to the 69th order of the eigen frequency. The eigen frequency obtained by the symmetric square wave modulation method is $\tilde{f}_e = 5050$ Hz, and the 69th order eigen frequency under this method is denoted as $\hat{f}_m = 348450$ Hz. The eigen frequency and the 69th order eigen frequency calculated by the SHD algorithm are respectively denoted as f_{e0} and f_{m0} .

The starting point of the iteration process is arbitrarily chosen within the interval $(69\hat{f}_e, 71\hat{f}_e)$, and is set as $f_0 = 356000$ Hz. During the iteration, the values at each point and the difference from the previous iteration point are as follows in Table II. The threshold for stopping the iteration is set as $\varepsilon_f = 0.01$ Hz.

From Table II, it can be seen that the SHD algorithm, even when starting from an arbitrarily chosen point, finds the 70th harmonic point of the eigen frequency f_e in just 4 steps of iteration. Compared to the conventional square wave feedback method that requires frequency sweeping, the proposed algorithm greatly saves on calculation resources. Moreover, the error produced by the iteration threshold ε_f occurs at the high-order eigen frequency $2kf_e$, so when calculating the eigen frequency f_{e0} and the length of the fiber-optic coil, this error is reduced by a factor of $2k$. This significantly enhances the measurement precision of fiber length. According to Table II, $f_{e0} = 5074.018$ Hz, and the measured fiber length L is:

$$L = \frac{c}{2n_{\text{eff}}f_{e0}} = 20303.724 \text{ m.} \quad (37)$$

And according to (32), fiber length error is calculated as:

$$\Delta L = \frac{\varepsilon_L}{1 + \varepsilon_L} L = 0.57 \text{ cm.} \quad (38)$$

The calculation in (37) is an example for that the SHD algorithm can obtain a high relative precision in fiber length. Furthermore, as shown in (38), the SHD algorithm obtains a low fiber length error under the determined parameters. It means that this algorithm can precisely detect tiny changes in fiber length after calibration.

In addition, for the giant IFOG, we also changed the frequency of the starting points to verify the convergence of SHD algorithm in any case. The starting points are set at 500 Hz intervals based on f_0 , covering a width of $2f_{e0}$. The experimental results are shown in Fig. 13. The red dotted line represents the variation of the second harmonic amplitude with frequency in one period for reference.

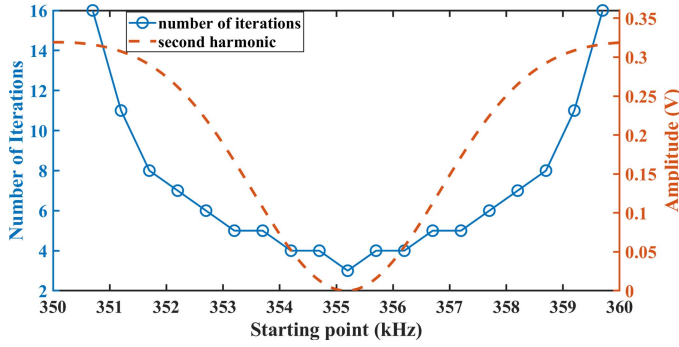


Fig. 13. Number of iterations corresponding to various frequency starting points of SHD algorithm is marked by blue circles. As a reference, the variation of the second harmonic amplitude with frequency is given by the red dotted line.

It can be seen that the SHD algorithm is convergent at any starting point in one period. However, as the starting point changes from the even-order eigen frequency point to the odd-order eigen frequency point, the number of iterations required is constantly increasing. The closer to the odd-order eigen frequency point, the faster the number of iterations increases. This is caused by the concavity of the second harmonic amplitude itself. Despite that the iteration speed depends on the chosen starting point, the process will ultimately converge with an error less than 0.01 Hz.

Compared with existing measurement methods, this algorithm has several advantages. For one thing, it is faster as it requires no frequency sweeping and can determine the eigen frequency only by measuring a few points. For another, it has relatively high precision. The measurement error at 70th order eigen frequency can be less than 0.01 Hz, indicating that the corresponding relative error can reach 28 ppb. Additionally, according to (32), the relative length error for fiber-optic coil over 20 km is less than 1 cm.

To further illustrate the role of the SHD algorithm in measuring eigen frequency, a comparative experiment is conducted to assess the impact of the fiber length and eigen frequency obtained on the overall system performance. A static observation is conducted, where the modulation frequencies are selected as follows: the 69th high-order eigen frequency $\tilde{f}_m = 348450$ Hz obtained by the conventional square wave feedback method, and the 69th high-order eigen frequency $f_{m0} = 350107$ Hz solved by the SHD algorithm.

The performance comparison between the two is based on the premise that the mean values of the data collected are consistent. Their Allan standard deviation curves are plotted and compared together, as shown in Fig. 14, and their performances are listed in Table III.

From Fig. 14 and Table III, it is evident that SHD algorithm has an enhancing effect on the ARW. Compared to high-order frequency modulation by conventional square wave feedback method, high-order frequency modulation solved by the SHD algorithm reduces the ARW by 4.5 times to $2.2 \times 10^{-5} \text{ }^\circ/\sqrt{\text{h}}$. The experimental results demonstrate that the SHD algorithm indeed improves the precision of the eigen frequency of the IFOG, thereby enhancing its sensitivity.

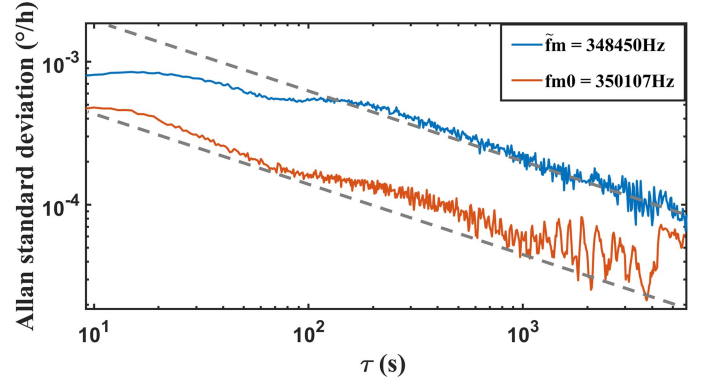


Fig. 14. The Allan standard deviation of IFOG when the modulation frequency is obtained by the conventional square wave feedback method and the proposed SHD algorithm.

TABLE III
PERFORMANCE COMPARISON BETWEEN CONVENTIONAL SQUARE WAVE FEEDBACK METHOD AND SHD ALGORITHM

Method	Obtained Modulation frequency(Hz)	Angular random walk($^\circ/\sqrt{\text{h}}$)
Conventional square wave feedback method	$\tilde{f}_m = 348450$	1.0×10^{-4}
Proposed SHD algorithm	$f_{m0} = 350107$	2.2×10^{-5}

V. DISCUSSION

As stated in Sections III and IV, the main competitive advantage of the proposed SHD algorithm lies in its ability to accurately determine the eigen frequency and length of the fiber-optic coil by measuring only a few points. This algorithm differs from the mainstream frequency sweeping methods and also offers much greater measurement precision than the conventional square wave feedback method. At the same time, this algorithm has also been applied to the IFOG. Experiments show that by obtaining the high-order eigen frequency through the SHD algorithm and using this as the modulation frequency, the ARW of giant IFOG can be improved.

For giant IFOGs, high frequency modulation is necessary due to the effect of TPN. However, when the modulation frequency deviates from the eigen frequency and its odd harmonics, it can lead to degradation of the performance of the IFOG. This is primarily caused by the phase error $\phi_{bs,ph}$ of the Rayleigh backscattering phase noise, as analyzed in Section II. By contrast, the phase error $\phi_{bs,I}$ caused by intensity noise of Rayleigh backscattering has a much smaller impact, which can be ignored. The phase error $\phi_{bs,ph}$ can be eliminated when f_m is set as f_e and its odd harmonics. Therefore, the setting of the modulation frequency depends on precise high-order eigen frequency measurement. This is of great significance for IFOGs in noise suppression, especially for giant ones, and puts forward a high requirement for the measurement precision of eigen frequency.

The introduction of the SHD algorithm is aimed at suppressing errors in modulation frequency. Section IV shows that the SHD algorithm greatly aids in measuring eigen frequency of fiber-optic coil simply and precisely. Meanwhile, as discussed

in Section III-C, when using the SHD algorithm, it's crucial to ensure (33) is satisfied. This is because the voltage resolution could lead to measurement errors. For this, directly measuring the minimum point of the second harmonic is not feasible. The smallest voltage V_e that the detection system can detect cannot exceed the mean value of the system circuit noise. At this point, the SHD algorithm becomes significantly important. It selects the surrogate function $A_{cat}(f)$ for the second harmonic as a quadratic function, ensuring the direction and speed of each iteration. Specifically, as discussed in Section IV, we need to converge ε_f to below 0.01 Hz. However, from Table II it can be seen that in the previous iteration step, ε_f only needs to converge to around 0.1 Hz. This successfully avoids the issue of insufficient voltage resolution.

During the experiment, the SHD algorithm always converges to the minimum point of the second harmonic, regardless of where the starting point is chosen. It is worth noting that the selection of the starting point does affect the convergence speed. This is determined by the characteristics of the second harmonic, whose concavity and convexity are not consistent. The strategy of selecting the starting point is not to be too far away from the even-order eigen frequency point, so that the steps of iteration can be reduced.

Furthermore, the application of the SHD algorithm is not limited to giant IFOG. It can be applied to any Sagnac interferometer. The only requirement is to measure the second harmonic amplitudes corresponding to a few modulation frequencies. This method is simple and easy to operate, making it suitable for both low-cost applications (such as inertial navigation) and high-precision measurement fields (such as fiber length measurement, temperature effect compensation, seismic measurement, etc.). SHD also has good compatibility, allowing it to be used in conjunction with other methods in terms of hardware facilities and signal processing to further enhance the performance of IFOGs.

VI. CONCLUSION

In this study, we first analyzed the phase noise induced by modulation frequency deviation. Based on the analysis, we proposed the second harmonic descent (SHD) algorithm. The introduction of surrogate function makes the proposed method converge quickly, thus using less computing resources. In this way the eigen frequency can be determined precisely, and a fairly precise estimate of the fiber length can also be obtained. To verify the effectiveness of the SHD algorithm, static tests have been carried out on an IFOG. By using the proposed SHD algorithm, the measurement precision of high-order eigen frequency can reach 0.01 Hz, and the corresponding relative error can reach 28 ppb. Compared with the conventional square wave feedback method, the proposed SHD algorithm can

improve ARW by 4.5 times. This proves the effectiveness of the SHD algorithm in suppressing the phase noise induced by the modulation frequency deviation. The SHD algorithm is simple and easy to implement, offering relatively high precision. It not only facilitates the low-cost application of IFOGs but also aids in high-precision measurements. Combined with other methods, the SHD algorithm can advance developments in fields such as inertial navigation and seismological research.

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